LESSON 5.7

Determine Surface Area of Prisms

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of rectangular and triangular prisms.

WHY THIS MATTERS Your class is painting the outside of different time capsule shapes to protect them from weather. Some capsules are rectangular prisms and others are triangular prisms (wedge-shaped). To buy the right amount of paint, you need to calculate the surface area of each shape! One capsule is a rectangular prism 7 inches long, 4 inches wide, and 3 inches tall. Another is a triangular prism whose triangle bases are 8 inches wide and 3 inches tall, with slant sides of 5 inches, and the prism is 9 inches long.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Surface area *Spanish: Área de superficie*	The total area of all the flat sides of a solid.	add every face
Rectangular prism *Spanish: Prisma rectangular*	A solid box shape with six flat rectangle sides.	3 congruent pairs
Triangular prism *Spanish: Prisma triangular*	A solid with two triangle ends and three flat rectangle sides.	Shaped like a tent or a wedge of cheese — 2 triangle ends + 3 rectangle sides
Face *Spanish: Cara*	One flat side of a solid shape.	top, side, end
Net *Spanish: Plantilla (desarrollo plano)*	A flat shape that folds up into a solid.	unfolded box
Lateral face *Spanish: Cara lateral*	The side faces of a solid, not the top or bottom.	On a triangular prism, the 3 rectangles that wrap around the sides are lateral faces

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A closed box is 6 cm × 4 cm × 3 cm.

1. Faces: 6 × 4 = 24, 6 × 3 = 18, 4 × 3 = 12.
2. Double each: 48, 36, 24.
3. Add: 48 + 36 + 24 = 108.

Answer: SA = 108 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a rectangular prism: 5 long, 4 wide, 3 tall. How many faces does it have?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The prism has _____ faces in _____ matching pairs.
- The three different face areas are _____ , _____ , and _____ .
- I doubled each face area because _____ .
- The surface area is _____ square units.

Word bank surface area | net | face | congruent | pair | prism | unfold | add | double | square units | top | side

WATCH OUT!

A common mistake

A common mistake in Surface Area of Prisms is confusing surface area with volume

LESSON 5.7

Determine Surface Area of Prisms

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Put the steps in order for finding the surface area of a triangular prism.

Groups: Steps in order (first → last)

_____ Find the area of one triangular base

_____ Multiply the triangle area by 2 (two bases)

_____ Find the area of each rectangular lateral face

_____ Add all three rectangle areas together

_____ Add the two triangle areas + all rectangle areas = SA

STEPS TO TRY

1. The cards describe finding a triangular prism's SA: triangles first, then rectangles, then a final total.
2. Order it like building a net: one triangle's area, double it, each rectangle's area, add rectangles, combine all.
3. Place 'Find the area of one triangular base' first — you need one triangle before you can double it.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A time capsule is a rectangular prism with l = 5 cm, w = 4 cm, h = 3 cm. What surface area needs paint?

(A) 47 cm²

(B) 60 cm²

(C) 94 cm²

(D) 94 cm³

STEPS TO TRY

1. You need surface area — the total of all 6 face areas of the 5 × 4 × 3 prism — not the space inside.
2. Use SA = 2lw + 2lh + 2wh: three different faces, each with a matching twin on the opposite side.
3. Find the three face areas first: 5×4 = 20, 5×3 = 15, and 4×3 = 12. Double each, then add the results.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

How many faces does a triangular prism have?

(A) 3

(B) 4

(C) 5

(D) 6

STEPS TO TRY

1. Picture a tent or a triangular chocolate box: two triangle ends plus rectangles wrapping around the sides.
2. Count the two triangular bases first, then count one rectangle for each side of the triangle.
3. A triangle has 3 sides, so count 3 rectangular faces. Then add the triangular ends to that count.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A wedge-shaped time capsule is a triangular prism. It has two triangular bases each with area 12 cm² and three rectangular faces with areas 30 cm², 40 cm², and 40 cm². What is the total surface area to paint?

(A) 110 cm²
(B) 122 cm²
(C) 134 cm²
(D) 134 cm³

STEPS TO TRY

1. All five face areas are given: two triangles of 12 cm² each, and rectangles of 30, 40, and 40 cm².
2. Total surface area = every face added once — count BOTH triangular bases, not just one.
3. Start with the triangles: 2 × 12 = 24 cm². Then add the three rectangle areas to that.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

Compare Capsule A (rectangular, SA = 248 in²) and Capsule C (triangular, SA = 184 in²). Even though they are similar in size, why does the rectangular prism need more paint?

SENTENCE STARTER The rectangular prism has ___ faces while the triangular prism has ___ faces. The rectangular prism's SA is ___ in² which is ___ more than the triangular prism's ___ in² because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help